

Issue 128 Six-Hour HPC Sprint

Proof-Carrying Technical Supplement

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Abstract

This supplement records a six-hour, proof-directed HPC sprint for Issue 128. Before using any new HPC result, a post-freeze D5 sidecar was integrated into the global error ledger and independently verified. It accepts 95 steps, rejects 94, and certifies the exact resource ratio $11791/2851 = 4.135741844966678\dots$. The production phase computes exact right-generator coefficients through complementary monolithic and 31-way stage-sharded paths. Every accepted artifact is canonical, SHA-256-bound, source-identified, and checked by a verifier that can reject malformed coverage, algebra, provenance, boundary, or resource arithmetic. No fivefold result is asserted.

1 Certified result and research gate

For the periodic 12×12 isotropic Heisenberg benchmark at $T = 1$ and global operator-norm tolerance 10^{-6} , the certified fourth-order Suzuki post-freeze result accepts $r = 95$ and rejects $r = 94$. With $G(r) = 30r + 1$ merged group exponentials, the exact resource ratio is

$$\frac{G(393)}{G(95)} = \frac{11791}{2851} = 4.135741844966678\dots$$

The exact global bounds are $9.90449140283245 \times 10^{-7}$ at 95 and $1.04680611656037 \times 10^{-6}$ at 94. The D4 and D5 sidecars contain 75,324 and 605,832 canonical Pauli terms, respectively, and both are independently checked for complete coverage and pairwise anticommutation within every group. The original ten-file package remains frozen at 97 steps and is not silently rewritten.

Fivefold arithmetic would require $r \leq 78$, but the present D4 contribution alone is already approximately 1.259084×10^{-6} there. Thus an exact D8 coefficient, even if zero, cannot close fivefold performance under the current nonnegative ledger. A new multiplier may be reported only after a complete global ledger and independent fast/deep verification succeed.

2 Exact sharding algorithm

For stages $V_j(t) = \exp(-ia_j t H_{c_j})$ in the fixed 31-stage formula, linearity gives a decomposition of the right generator into 31 contributions. Each array task starts from $a_j H_{c_j}$ and conjugates it through all later stages, truncating only the formal series degree at eight. Coefficients live exactly in $\mathbb{Q}[\alpha]/(\alpha^3 - 4)$; Pauli operators are canonicalized under the 2×2 translation cell before serialization.

The shard format does not expose process-local registry bit indices. Every Pauli string is encoded by physical integer coordinates and labels. Canonical JSON and gzip with fixed metadata make repeated output byte-deterministic. The manifest records stage, order, timestamps, term counts, peak RSS, source commit, dirty flag, and SHA-256. The reducer rejects a missing stage, wrong formula/order, mixed source state, corrupt hash, or duplicate stage. It also requires forward and reverse exact reduction to produce identical maps.

For a cubic coefficient $c = a_0 + a_1\alpha + a_2\alpha^2$, an outward rational interval for $\alpha = \sqrt[3]{4}$ encloses c . Summing the larger endpoint magnitudes yields a rigorous Pauli- ℓ_1 upper bound. This bound is a screening ingredient; it is not by itself the global $r = 78$ certificate.

3 Execution contract

The original D8 array source is commit `e65c95f4ae76dc1a78e81715cac21cfd7a81dbf8`; the cancellation-aware monolithic and order-six stage source is `eade9d9e0bc56841ee620e83582382d902c0d625`; the initial exact E7 word-sharded lane is commit `23dbe9532929d81c8cbb366e12a3d86c03bd4e9e`, with the 272- and 544-way fanout wrappers at commits `ab387bf3...` and `88a4f2ed...`. Relevant source-file digests were checked byte-for-byte between the local and remote trees. Jobs run on the `xhacnormalb` CPU partition under account `giggleliu` and QOS `user_acamtw70yu`. GPU partitions are intentionally excluded because the workload is sparse Python integer/rational algebra.

Role	Slurm job	Per-task allocation	Limit
Exact D8 array (31 stages)	23044178	17 CPU, 64 GiB	4:30
Exact D8 monolithic fallback	23044280	63 CPU, 240 GiB	4:15
Exact D6 monolithic path	23044494	9 CPU, 32 GiB	3:30
Exact D6 array (31 stages)	23044609	9 CPU, 32 GiB	2:30
D6 array reducer	23044651	26 CPU, 96 GiB	1:00
Direct exact D6 identity	23044709	17 CPU, 64 GiB	1:45
Exact E7 word array (16 shards)	23044736	17 CPU, 64 GiB	1:00
E7-to-D6 exact reducer	23044752	26 CPU, 96 GiB	0:35
E7 CPU fanout (272 shards)	23044754	17 processes/cell	0:45
Fanout E7-to-D6 reducer	23044907	26 CPU, 96 GiB	0:40
E7 fine fanout (544 shards)	23044779	17 processes/cell	0:40
Fine-fanout E7-to-D6 reducer	23044906	26 CPU, 96 GiB	0:45
E7 ultra-fine fanout (1,088 shards)	23044815	17 processes/cell	0:30
Ultra-fine E7-to-D6 reducer	23044816	26 CPU, 96 GiB	0:24

Table 1: Principal production work. Reducers start only after all parent array cells succeed. The D6 reducer has an 08:32 CST deadline and the E7 reducer an 08:34 deadline.

The array first passed `sbatch -test-only`. A four-cell remote smoke then completed the lightest stages and produced matching deterministic artifacts. The partition’s 3931 MiB-per-CPU memory rule was discovered by the test-only gate and reflected in all allocations. Once production started, the voluntary array throttle was raised from 16 to 31; Slurm and QOS retained authority over placement and admission.

4 Verification evidence

The final local regression gate reported 112 passing tests and 11 deselected slow tests in 175.21 seconds. The D5-integrated fast verifier returned `valid=true`; the deep verifier regenerated every D4/D5 coefficient from the 31-stage formula, returned `deep_proof_regenerated=true`, and completed in 167.62 seconds with 1.53 GB peak RSS. A detached clean worktree repeated the fast verifier, focused certificate tests, and every delivery SHA-256 check.

The independent remote verification job is recorded without hiding a mixed outcome: its deep verifier passed, while the encompassing `pytest` command returned nonzero. Five failures were caused by the synchronized remote tree lacking usable Git metadata; one pre-existing slow symplectic discovery test returned a different numeric value. This remote `pytest` result is not called “all green” and is not used to replace the successful local certificate gate.

The exact-D8 array provides a useful scale check. Stage 25 completed in 7349.722 seconds with 17,211,714 degree-eight terms and 52.15 GB peak RSS; its output digest is `c64c0729a1ef...a95a4eb3cf7b`; the full digest is retained in the machine-readable sprint log. The order-six fallback uses the same stage-shard contract but truncates at D6. Its reducer requires all 31 manifests, verifies each payload digest, and requires forward and reverse exact sums to agree. A compact parent manifest binds every source shard and the reducer implementation before the exact D6 sidecar can enter the main certificate.

An additional cross-order replay job compared the degree-six maps embedded in the independently generated order-six and order-eight shards for stages 25–30. All six exact maps agreed term by term, covering 710,298, 446,094, 150,858, 2,028, 48, and zero terms, respectively. This check compares canonical Pauli keys and exact cubic-field coefficients rather than only a reported norm.

The matching-order array is a separate discovery lane. Its JSON records `trusted=false` and states that its purpose is discovery ranking. A favorable ranking can nominate a formula for exact reconstruction but cannot enter the certificate

directly.

The 25 heavy D8 array cells remained in `RUNNING` state beyond 4:15:00 and ultimately reached `TIMEOUT` after 4:30:11–4:30:13 of observed wall time. The other six cells completed. Thus the requested 4.25-hour production floor is supported by `sacct`, not inferred from a requested time limit. Because the full array was not successful, Slurm cancelled its `afterok` reducer; no partial sum was reported globally.

At 07:13 CST a complementary 16-way E7 word array, job 23044736, began from byte-identical local and remote sources. It partitions all 16,380 degree-seven free words into disjoint index intervals. Dependent job 23044752 validates complete interval coverage, source and payload hashes, and forward/reverse exact reduction before forming $D_6 = 7E_7 + \frac{2}{3} \text{ad}_A^2(E_5)$. Its result remains provisional until the complete artifact and verifier gates pass.

The first minute of the 16-way lane exposed load imbalance: three contiguous blocks finished in roughly 43–45 seconds with zero E7 contribution, while the other blocks entered expensive words. Per-process RSS was only about 165–203 MiB despite a 64 GiB allocation. A CPU-saturating fallback therefore launched as job 23044754: each of 16 array cells uses `srun` to run 17 single-core evaluators, partitioning the same word set into 272 disjoint intervals. All 16 cells began within 11 seconds and all 272 running manifests were observed. Job 23044755 reduces only that lane and retains the same coverage, hash, bidirectional-sum, and exact-identity gates.

After about five minutes the 272-way lane had 155 complete shards, 117 still running, and no failures, with the unfinished set concentrated in expensive word ranges. A second independent lane, job 23044779, therefore halves the range size again to 544 shards across 32 array cells and 544 active Python evaluators. All cells entered `RUNNING` within 11 seconds and emitted their manifests. Reducer 23044780 reads only this lane. If both reductions finish, their exact D6 maps must agree term by term before either can support a certificate.

At roughly four minutes the 544-way lane had completed 383 shards with no failure; completed high-cost shards had reached about 3 GiB peak RSS. A final 1,088-way lane, job 23044815, reduced each range to about 15 words and started all 64 cells (1,088 evaluators). Reducer 23044816 is directory-isolated from the other lanes. The first complete result wins time-to-answer, while any additional complete result supplies an exact cross-partition comparison.

A consolidated 07:28:33 CST snapshot recorded 1,748 allocated CPUs in running jobs for the account. The 272-, 544-, and 1,088-way E7 lanes had completed 242, 482, and 567 shards, respectively, with zero failed shards. Their maximum completed-shard RSS values were about 6.41, 4.15, and 1.33 GiB. Thus the finer partition increased active evaluator concurrency while reducing individual memory and tail exposure; the reported figure is scheduler state, not a sum of requested limits.

Before either fanout reducer became eligible, deployment preflight found that its Slurm wrapper still passed a literal shard count of 16. The wrapper was changed to consume the exported `ISSUE128_SHARD_COUNT` while retaining 16 as the default, passed `bash -n`, was synchronized byte-for-byte, and was inspected remotely. Jobs 23044755 and 23044780 remained dependency-pending throughout the correction and were submitted with counts 272 and 544, respectively; no reduction had begun with the stale parameter.

Slurm snapshots a batch script at submission. Consequently, the already submitted 544-way reducer 23044780 still started with the old literal count and failed closed in two seconds when shard 0 declared 544 rather than 16; its output directory remained empty. Replacement 23044906 was submitted from the corrected wrapper and immediately ran with a 45-minute limit. The still-pending old 272 reducer 23044755 was cancelled before execution and replaced by 23044907 with the original parent dependency. The 1,088-way reducer 23044816 had been submitted after the correction and passed the metadata gate.

5 Final adjudication protocol

The outcome will be classified as follows.

1. If all 31 shards and exact reduction succeed, insert the resulting D8 upper bound into a complete $r = 78$ ledger and rerun fast/deep checks.
2. If any task reaches OOM or walltime, retain successful shards and scheduler evidence, retry only missing cells if time remains, and make no inference from a partial sum.
3. If the complete ledger is at most 10^{-6} , prepare a new independently bound certificate before describing a fivefold result.
4. Otherwise publish the strict negative result or remaining numerical gap, while keeping the certified 4.135742-fold result unchanged.

6 Observed final outcome

The original D8 array ended with six completed and 25 timed-out cells; each heavy cell recorded 4:30:11–4:30:13 of actual elapsed time. Its reducer did not run. The monolithic D8 fallback timed out at 4:15:29. The monolithic D6 path timed out at 3:30:05, and its 31-way stage array ended with 13 completed and 18 timed-out cells. Neither path emitted a complete global coefficient map.

The CPU-saturating E7 parents did close completely: the 272-, 544-, and 1,088-shard lanes completed all 16, 32, and 64 Slurm cells, respectively, with complete word coverage and no failed shard manifest. Their subsequent D6 reducers reached walltime at 40:07, 45:06, and 24:21. Because output is written atomically, no partial `degree-6.json.gz` was left behind. The dependent cross-lane comparisons and certificate integration therefore did not run. The original 16-way E7 lane ended with four completed and 12 timed-out cells. The serial direct-identity lane timed out at 1:39:06 after reaching word 1,792 of 16,380. At 08:24:40 CST the user queue was empty; no sprint job remained beyond the bounded execution window.

The final local regression gate was 112 passed and 11 deselected in 175.21 seconds. Operationally, the sprint proved the requested 4.25-hour production floor, exercised up to 1,748 running CPU allocations, and produced complete hash-bound E7 word partitions, but did not close a new D6 or D8 reduction within six hours. This is recorded as a strict negative operational result, not converted into a numerical claim.

Claim status. The currently accepted result is the independently verified 4.1357418449-fold D5-integrated certificate. The D6/D8 work is research evidence only until its artifacts, exact ledger, adjacent-step boundary, hashes, and deep regeneration all pass.